Unit - I
INTRODUCTION

What is Green Building, Why to go for Green Building, Benefits of Green Buildings, Green Building Materials and Equipment in India, What are key Requisites for Constructing a Green Building, Important Sustainable features for Green Building,

1.1 Green Building

A 'green' building is a building that, in its design, construction or operation, reduces or eliminates negative impacts, and can create positive impacts, on our climate and natural environment. Green buildings preserve precious natural resources and improve our quality of life.

Sustainable Development

- "Sustainable development is development that meets the needs of the present, without compromising the ability of future generations to meet their own needs."
- Sustainable development is about finding better ways of doing things, both for the future and the present.

1.2 Why to go for Green Buildings

The global construction and building sector is one of the largest contributors to climate change and resource depletion. Continuing down the traditional path poses severe risks to our economies, health, and climate stability.

1. The Climate Crisis & Carbon Budgets

The building sector is responsible for nearly 39% of global energy-related carbon emissions. This is split into two categories:

- Operational Emissions (28%): The energy used to heat, cool, and light buildings daily.
- Embodied Carbon (11%): The carbon footprint of manufacturing, transporting, and assembling materials like concrete and steel.

We cannot hit global net-zero climate targets without decarbonizing buildings. Green buildings are a necessity to keep global temperature increases within safe limits and prevent catastrophic climate events.

2. Escalating Resource Scarcity

Traditional construction is incredibly resource-intensive, straining finite global supplies.

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- Water Stress: Global water scarcity is rising rapidly. Traditional buildings put massive pressure on municipal water supplies. Green buildings use rainwater harvesting and greywater recycling to significantly cut demand.
- Material Depletion**:** Construction consumes billions of tons of raw materials annually. Green buildings demand the use of recycled, rapidly renewable, and circular materials, preventing the total depletion of natural resources.

3. Skyrocketing Energy Costs & Grid Failures

The global energy grid is under immense stress due to rising temperatures and increased cooling demands.

- Traditional, poorly insulated buildings act like energy sieves, putting massive strain on electrical grids during peak summer and winter months.
- Green buildings utilize passive design—using smart insulation, strategic window placement, and natural ventilation—to stay comfortable without relying heavily on the grid. They reduce the risk of rolling blackouts and protect occupants from skyrocketing energy prices.

4. Regulatory and Policy Mandates

Governments worldwide are rapidly shifting green regulations from "encouraged" to "legally mandated."

- Many major global cities are enforcing strict carbon limits on large buildings, introducing heavy fines for properties that exceed emission caps.
- Financial institutions are increasingly tying commercial loans and investment capital to ESG (Environmental, Social, and Governance) scores.

Soon, traditional buildings will become stranded assets—unsellable, unrentable, and illegal to operate under modern environmental laws. Building green is now a necessity for basic financial survival in the real estate market.

5. Urban Heat Islands & Public Health

As cities expand with concrete and asphalt, they trap heat, creating "urban heat islands" that make cities significantly hotter than surrounding rural areas.

- Green buildings counter this by integrating green roofs (vegetation), living walls, and cool, reflective pavements. Without them, our growing urban centres' will face unbearable, dangerous heat levels during summer months.

1.3 Benefits of Green Building

Going for Green Buildings have tremendous benefits, both tangible and intangible.

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The most tangible benefit is in reduction of operating energy and water costs right from day one during the entire life cycle of the building

Green Buildings offer a range of benefits like:

- 30% to 40% reduction in operation cost
- Green Corporate image
- Health and safety of building occupants
- Enhance occupant comfort
- Improve productivity of occupants
- Imbibe best operational practices from day one
- Incorporates latest techniques and technologies

Environmental Benefits

- Reduces impact of natural resource consumption
- Develops long-term resource/energy strategies that are practical in use

Economic and Social Benefits

- Reduces reliance on public utilities
- Increases the value of a home and increases resale interest
- Increases home durability
- Encourages sustainable building practices
- Reduce the impact on natural resource consumption and leave for future generations

Health and Safety Benefits

- Reduces exposure to allergens, carcinogens, pollutants
- Enhances occupant comfort and health
- Improve productivity of occupants

Community Benefits

- Reduces strain on local infrastructures and improves quality of life
- Provides homeowner pride and drives values for community members

Competitive Costs

- Integrated Green Design allows higher benefits at lower cost by achieving synergies between disciplines and between technologies

Operating and Tax Costs

- Utility costs significantly lower
- Life-cycle economic performance optimized
- Owner access to tax credits

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Increases Building Valuation and ROI

- Quantifying financial benefit in terms of return on investment (ROI) instead of payback time demonstrates true financial advantage

Faster environmental clearances for projects

- In a major impetus to the Green Buildings movement in India, Government of Maharashtra has in-principle approved Priority Status for "Faster environmental Clearance for IGBC registered projects" from the Environment Department and other Municipalities.

1.4 Green Building Materials and Equipment in India

1.4.1 Green Building Materials

i. Fly-ash bricks & Fly-ash based blocks

- Manufactured using coal-fired power plant waste (fly ash). Lighter, uniform, good compressive strength, lower water absorption and reduced top-soil depletion vs traditional clay bricks. Widely adopted in India for walls, pavements.

ii. Pozzolana Cement

- Portland Pozzolana cement is integrated cement which is formed by synthesising OPC cement with pozzolanic materials in a certain proportion. It is commonly known as PPC cement.
- A pozzolanic material can be a natural or artificial which contains silica and aluminous in a reactive form. These materials usually don't possess any cementitious properties, but when it is mixed with water or moisture or lime to undergo reaction with calcium hydroxide to form compounds possessing cement properties.

iii. AAC (Autoclaved Aerated Concrete) blocks

- Lightweight, insulating precast blocks — reduce wall weight and cooling loads, speed construction. Popular in mid-to-high rise and commercial projects.

iv. Compressed Stabilized Earth Blocks (CSEB) & Stabilized Mud Blocks

- Locally-made earth mixed with stabilizer (cement/lime) and compressed. Low embodied energy, good thermal mass, low cost where suitable soils exist. Good for vernacular and low-rise construction.

v. Bamboo

- Bamboo as a building material is used for the construction of scaffolding, bridges and structures, houses.
- Bamboos are one of the fastest-growing plants in the world and their growth is three times faster than most other species of plants.
- They are renewable and extremely versatile resource with multi-purpose usage.

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- Among many uses of bamboo, Housing is one of the major areas applications
- especially in the wake of residential shortages around the globe.

vi. Recycled / Secondary Steel & Rebar with high recycled content

- Steel with high recycled content reduces embodied energy vs primary steel. Use from reputable manufacturers and check mill/test certificates.

vii. Low-VOC paints, adhesives & finishes

- Low volatile organic compound products reduce indoor air pollution and occupant health risks. Many manufacturers in India offer low/no-VOC lines.

viii. Insulation materials (mineral wool, EPS, PIR, sheep-wool alternatives)

- Proper insulation for walls/roof reduces heating/cooling loads. Choose fire-rated and moisture-resistant options as per application.

ix. High-performance glazing & fenestration

- Double glazing, low-e coatings, proper frame thermal breaks reduce heat gain and improve daylighting control.

x. Green roof / vegetated roof systems and cool roofing membranes

- Reduce urban heat island effect, add insulation, manage stormwater. Cool reflective roof coatings are widely used when green roofs aren't practical.

xi. Permeable paving & recycled aggregates

- Permeable pavers reduce runoff; crushed/recycled concrete aggregates lower demand for virgin aggregates.

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Bamboo can be shaped in the following ways

Squared cross-section can be obtained by compressing the growing stalk of bamboo within a square section.

Arch shapes of bamboo can also be created by compressing the bamboo's growth into the desired shape. This would cost lesser than it would to get the same form with normal timber.

Curved and Flat shapes of bamboo are achieved through traditional techniques like applying heat and pressure.

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The various advantages of bamboo are as mentioned below:

- Tensile strength: Bamboo has higher tensile strength than steel because its fibers run axially.
- Fire Resistance: Capability of bamboo to resist fire is very high and it can withstand temperature up to 400^0 C. This is due to the presence of high value of silicate acid and water.
- Elasticity: Bamboo is widely preferred in earthquake prone regions due to its elastic features.
- Weight of bamboo: Bamboos due to their low weight are easily displaced or installed making it very easier for transportation and construction.
- Unlike other building materials like cement and asbestos, bamboo poses no danger to health.
- They are cost effective and easy to use.
- They are especially in great demand in earthquake prone areas.

Eucalyptus:

Eucalyptus wood, a fast-growing hardwood, is sustainable because new growth only takes a few years to mature.

Building with eucalyptus wood is desirable because, unlike other timbers, it has very little taper over long lengths. This quality makes it more consistent and efficient. There are more timber and less waste. And generally speaking (each Eucalyptus species is slightly different) it is harder that most North American woods.

Because eucalyptus wood is sustainable, it is now being used as an alternative to mahogany and Brazilian cherry. Both of these species are endangered, slow-growing and comparatively expensive. Building with eucalyptus wood is quickly becoming a natural choice.

Pressure treated eucalyptus poles have at least a 20-year ground burial rate.

Poplar Wood:

Wood from the yellow poplar tree is often used in light construction, such as in single-family homes or storage sheds. Wood from the yellow poplar tree is also used in kitchen cabinets and doors. Since wood from the yellow poplar tree can be easily machine-worked, painted and stained, this wood is an excellent choice for furniture making as well.

Rubberwood:

- Both eco-friendly and cost effective, rubber wood is ideal for furniture, flooring, ceiling, paneling and interiors
- Treated rubber wood possess advantages like
- Excellent stability
- Excellent machining properties
- Good strength properties
- Light color; easy to finish in any color

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- Good glaring properties
- Attractive grain structure
- Eco-friendly
- Up to 40% cheaper than other hardwoods

Linoleum:

Linoleum, also called Lino, is a floor covering made from materials such as solidified linseed oil (linoxyn), pine rosin, ground cork dust, wood flour, and mineral fillers such as calcium carbonate, most commonly on a burlap or canvas backing. Pigments are often added to the materials to create the desired colour finish.

Rammed Earth:

Rammed earth walls are constructed by ramming a mixture of selected aggregates, including gravel, sand, silt and a small amount of clay, into place between flat panels called formwork.

Traditional technology repeatedly rammed the end of a wooden pole into the earth mixture to compress it. Modern technology replaces the pole with a mechanical ram.

The many advantages of building with rammed earth include superior thermal mass, temperature and noise control, strength and durability, low maintenance, fire proofing, load bearing and pest deterrence, as well as its beauty and the pleasure of building with a natural and environmentally sound material.

Stabilized Mud:

The input of soil stabilization allowed people to build higher with thinner walls, which have a much better compressive strength and water resistance. With cement stabilization, the blocks must be cured for four weeks after manufacturing. After this, they can dry freely and be used like common bricks with a soil cement stabilized mortar.

Ideally, the production is made on the site itself or in the nearby area. Thus, it will save the transportation, fuel, time and money.

Well-designed Compressed Stabilized Earth Blocks (CSEB) houses can withstand, with a minimum of maintenance, heavy rains, snowfall or frost without being damaged. The strength and durability have been proven since half a century.

Adobe bricks:

Adobe bricks (mud bricks) are made of earth with a fairly high clay content and straw. The produced earth mix is cast in open moulds onto the ground and then left to dry out. Adobe bricks are only sun-dried, not kiln-fired. When used for construction they are laid up into a wall using an earth mortar. Before drying out, the finished walls are smoothed down. Often a clay render is applied as a surface coating.

As with other forms of earth construction, adobe bricks are a fireproof, durable yet biodegradable, non-toxic building material which provides sufficient thermal mass to buildings

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to ensure excellent thermal performance. Other benefits include low sound transmission levels through walls and a general feeling of solidity and security.

1.4.2 Green Building Equipment in India

i. *Solar Photovoltaic (rooftop) Systems*- Onsite renewable energy systems, particularly rooftop solar photovoltaic installations, help reduce dependence on the electricity grid, lower a building's carbon footprint, and contribute to sustainable energy generation. In many Indian cities, these systems offer attractive payback periods due to rising electricity tariffs and abundant solar resources. Furthermore, the Ministry of New and Renewable Energy (MNRE), Government of India, along with various state governments, supports rooftop solar adoption through incentives, subsidies, and net-metering policies, enabling consumers to export surplus electricity to the grid and receive credits against their energy consumption.

ii. *Solar Water Heaters (SWH)*- Solar water heating is a mature and widely adopted technology in India for meeting domestic, commercial, and institutional hot-water requirements. It significantly reduces electricity consumption by replacing or minimizing the use of conventional electric geysers, resulting in lower energy costs and reduced greenhouse gas emissions. The technology is reliable, cost-effective, and well-suited to India's climatic conditions, with numerous MNRE-listed manufacturers and vendors providing certified systems, installation, and maintenance services across the country.

iii. *Efficient HVAC: VRF/VRV, Heat Pumps, High-SEER Equipment*- HVAC systems are often the largest consumers of energy in commercial buildings; therefore, specifying high-efficiency equipment, Variable Refrigerant Flow (VRF) systems, and heat-recovery technologies wherever applicable can significantly reduce energy consumption and operating costs. The incorporation of thermostatic controls, programmable thermostats, and zoning strategies further enhances efficiency by enabling precise temperature regulation and conditioning only occupied spaces, thereby improving occupant comfort while minimizing unnecessary energy use.

iv. *LED Lighting & Smart Lighting Controls (occupancy sensors, daylight dimming)*- LED lighting systems can reduce lighting energy consumption by approximately 60–80% compared to traditional incandescent lamps and offer significant energy savings over older fluorescent lighting technologies. In addition to their higher efficiency, LEDs provide longer service life, lower maintenance requirements, and improved lighting quality. Further energy savings can be achieved by integrating occupancy sensors, motion detectors, daylight harvesting controls, and automatic dimming systems, which ensure that lighting is used only when and where it is needed while maximizing the utilization of available natural daylight.

v. *Building Management System (BMS) / Building Automation System (BAS)*- It integrates and centrally control building services such as HVAC, lighting, pumps, elevators, and renewable energy systems to optimize overall building performance. These systems continuously monitor energy consumption, equipment operation, and environmental conditions, enabling automated control strategies that improve efficiency, occupant comfort, and maintenance management. By providing real-time data, fault detection,

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and energy analytics, BMS/BAS solutions help reduce operational costs and energy wastage, making them particularly valuable and often essential for larger commercial, institutional, and industrial projects seeking to maximize long-term operational savings and sustainability performance.

vi. *Variable Frequency Drives (VFDs) for pumps & fans*- Variable Frequency Drives (VFDs) for Pumps and Fans enable motor speeds to be adjusted according to actual system demand rather than operating continuously at full speed. By matching the output of pumps and fans to varying load requirements, VFDs significantly reduce energy consumption, improve process control, and minimize mechanical wear and maintenance costs. They are widely recognized as a standard energy-efficiency measure for pumping systems, cooling towers, and central air-handling unit (AHU) fans, offering substantial operational savings while enhancing overall system performance and reliability.

vii. *Rainwater harvesting (RWH) & Groundwater recharge systems*- Rainwater Harvesting (RWH) is a critical water conservation measure, particularly in water-stressed Indian cities where increasing urbanization and groundwater depletion have intensified water scarcity challenges. Many state governments and urban local bodies have made RWH mandatory through municipal building regulations and development bylaws. Rainwater harvesting systems can range from simple groundwater recharge pits and percolation wells to advanced rooftop collection, storage, filtration, and reuse systems integrated into building infrastructure. These systems help reduce dependence on municipal water supplies, enhance groundwater recharge, and improve water security. To ensure compliance and effectiveness, several cities have initiated inspections and enforcement actions, including monitoring non-functional or poorly maintained RWH installations and implementing penalties where mandated systems are not operational.

viii. *Sewage Treatment Plants (STP) / Decentralized wastewater treatment & Greywater recycling*- Sewage Treatment Plants (STPs), decentralized wastewater treatment systems, and greywater recycling systems play a vital role in sustainable water management by enabling the treatment and reuse of wastewater within a building or campus. Treated wastewater can be effectively reused for non-potable applications such as toilet flushing, landscape irrigation, cooling tower makeup, and cleaning purposes, significantly reducing the demand for fresh potable water. Decentralized treatment systems are particularly suitable for residential complexes, institutional campuses, commercial buildings, and townships, where wastewater can be managed close to its source. The adoption of greywater recycling and wastewater reuse not only conserves water resources but also reduces the burden on municipal sewerage infrastructure. In India, several national and state-level manuals, standards, and guidelines have been developed to support the planning, design, operation, and maintenance of greywater management and wastewater reuse systems, promoting their integration into sustainable and green building practices.

ix. *Efficient pumps, water-efficient fixtures (low flow taps, dual-flush Water Closets)*- Efficient pumps and water-efficient fixtures, such as low-flow taps, aerators, sensor-based faucets, low-flow showerheads, and dual-flush water closets (WCs), are effective measures for reducing overall water consumption in buildings. By minimizing water

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demand, these systems also reduce the energy required for water pumping, treatment, and distribution, thereby contributing to both water and energy conservation. High-efficiency pumps further optimize system performance by delivering the required flow with lower power consumption. The Bureau of Indian Standards (BIS), along with green building rating systems such as IGBC and GRIHA, specifies performance criteria and benchmarks for water-efficient fixtures and pumping systems, encouraging their adoption as part of sustainable building design and operation.

x. *Energy-efficient elevators and regenerative drives*- Energy-efficient elevators and regenerative drive systems contribute to reducing the overall energy consumption of buildings by optimizing elevator operation and recovering energy that would otherwise be wasted. Regenerative drives capture the electrical energy generated when an elevator car travels downward with a heavy load or upward with a light load and feed this energy back into the building's electrical system for use by other equipment. In addition to regenerative technology, modern energy-efficient elevators incorporate features such as variable voltage variable frequency (VVVF) drives, LED cabin lighting, automatic standby modes, and intelligent dispatch controls, which further enhance energy performance. These systems are particularly beneficial in high-rise residential, commercial, and institutional buildings, where elevator usage is significant and energy savings can be substantial over the building's lifecycle.

1.5 Key Requisites for Constructing a Green Building

The key requisites for constructing a *Green Building* are the planning, design, construction, and operation practices that minimize environmental impact while ensuring occupant comfort and resource efficiency.

1. Sustainable Site Selection and Planning

- Choose sites with minimal ecological disturbance.
- Avoid construction on wetlands, forests, and fertile agricultural land.
- Ensure easy access to public transportation.
- Preserve existing vegetation and natural drainage patterns.
- Reduce heat island effect through landscaping and reflective surfaces.

Criteria for Site Selection

Do not develop buildings on portions of sites that meet any of the following on a site

a. Prime farmland
b. Previously under-developed land whose elevation is lower than 1.5m above the elevation of the 100yrs flood frequency.
c. Land specifically identified as habitat for any species on tranced or endangered list.
d. Land within 30m of any wetland and isolated wetlands or areas of special concern identified state or central rule
e. Previously under developed land that is within 20m of any water-body such as lakes, rivers and reservoirs.

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f. Site shall not be located near to solid waste disposal sites, oxidation ponds and odour producing industries.
g. The site location shall be close to suburban transportation route, and is within walking, biking, or bus distance to medical clinics, schools, parks, temples, shopping, parks and other recreational spots.
h. The site location should demonstrate cultural diversity consistent with that statewide.
i. The site shall not be located on brownfield which is a place of contamination. With hazardous substance or pollutants
j. The site location should not spoil the existing land scape and protect it from degradation during the process of construction.
k. The site soil should be good in terms of soil bearing capacity, compatibility and soil permeability

2. Energy-Efficient Building Design

- Orient the building to maximize natural lighting and ventilation.
- Use passive solar design principles.
- Optimize building envelope (walls, roofs, windows) for thermal performance.
- Install high-performance glazing and insulation.
- Incorporate energy-efficient HVAC systems.

3. Renewable Energy Integration

- Install solar photovoltaic (PV) panels.
- Use solar water heating systems.
- Consider wind energy or other renewable sources where feasible.
- Implement battery storage systems for improved energy management.

4. Water Conservation Measures

- Rainwater harvesting systems.
- Low-flow fixtures and water-efficient appliances.
- Greywater recycling systems.
- Efficient irrigation methods such as drip irrigation.
- Water metering and leak detection systems.

5. Sustainable Building Materials

- Use locally sourced materials to reduce transportation emissions.
- Select recycled and recyclable materials.
- Employ rapidly renewable materials such as bamboo.
- Use low embodied-energy materials.
- Avoid toxic and high-VOC (Volatile Organic Compound) materials.

6. Efficient Waste Management

- Segregate construction waste at source.

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- Reuse and recycle demolition and construction debris.
- Provide facilities for occupant waste segregation.
- Minimize landfill disposal.

7. Indoor Environmental Quality (IEQ)

- Ensure adequate fresh air ventilation.
- Maximize daylight penetration.
- Control indoor pollutants.
- Use low-VOC paints, adhesives, and finishes.
- Maintain thermal, visual, and acoustic comfort.

8. Green Landscaping

- Plant native and drought-resistant species.
- Develop green roofs and vertical gardens where feasible.
- Create shaded outdoor spaces.
- Enhance biodiversity around the site.

9. Smart Building Management

- Install Building Management Systems (BMS).
- Use occupancy sensors and automated lighting controls.
- Monitor energy and water consumption continuously.
- Implement predictive maintenance strategies.

10. Climate-Responsive Design

- Design according to local climate conditions.
- Utilize natural cooling and heating techniques.
- Provide protection against extreme weather events.
- Improve resilience to climate change impacts.

11. Life-Cycle Cost Consideration

- Evaluate materials and systems based on life-cycle cost rather than initial cost.
- Consider operation, maintenance, replacement, and disposal costs.
- Aim for long-term economic and environmental benefits.

12. Compliance with Green Building Rating Systems

Common certification systems include:

- Indian Green Building Council (IGBC)
- Green Rating for Integrated Habitat Assessment (GRIHA)
- Leadership in Energy and Environmental Design (LEED)

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1.6 Important Sustainable features for Green Building

These include

- Efficient use of energy, water and other resources
- Use of renewable energy, such as solar energy
- Pollution and waste reduction measures, and the enabling of re-use and recycling
- Good indoor environmental air quality
- Use of materials that are non-toxic and sustainable
- Consideration of the environment in design, construction and operation
- Consideration of the quality of life of occupants in design, construction and operation
- A design that enables adaptation to a changing environment

Energy Efficiency

- Passive design strategies can dramatically affect building energy performance.
- Develop strategies to provide natural lighting.
- Install high-efficiency lighting systems with advanced lighting controls.
- Use a properly sized and energy-efficient heat/cooling system in conjunction with a thermally efficient building shell.
- Minimize the electric loads from lighting, equipment, and appliances
- Consider alternative energy sources such as photo voltaic and fuel cells

Materials Efficiency

- Select sustainable construction materials and products by evaluating several characteristics
- Use dimensional planning and other material efficiency strategies.
- Reuse and recycle construction and demolition materials.
- Require plans for managing materials through deconstruction, demolition, and construction.
- Design with adequate space to facilitate recycling collection and to incorporate a solid waste management program that prevents waste generation.

Water Efficiency

- Design for dual plumbing to use recycled water for toilet flushing or a gray water system that recovers rainwater or other non-potable water.
- Minimize wastewater by using ultra-low-flush toilets, low-flow shower heads, and other water conserving fixtures.
- Use recirculating systems for centralized hot water distribution.
- Install point-of-use hot water heating systems for more distant locations.
- Use state-of-the-art irrigation controllers and self-closing nozzles on hoses.

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Occupant Health and Safety

- Choose construction materials and interior finish products with zero or low emissions to improve indoor air quality.
- Many building materials and cleaning/maintenance products emit toxic gases, such as volatile organic compounds (VOC) and formaldehyde. These gases can have a detrimental impact on occupants' health and productivity.
- Provide adequate ventilation and a high-efficiency, in-duct filtration system.
- Prevent indoor microbial contamination through selection of materials resistant to microbial growth, provide effective drainage from the roof and surrounding landscape, install adequate ventilation in bathrooms, allow proper drainage of air-conditioning coils, and design other building systems to control humidity.

Building Operation and Maintenance

- Green building measures cannot achieve their goals unless they work as intended.
- Over time, building performance can be assured through measurement, adjustment, and upgrading.
- Proper maintenance ensures that a building continues to perform as designed and commissioned.

Any building can be a green building, whether it’s a home, an office, a school, a hospital, a community centre, or any other type of structure, provided it includes features listed above. However, it is worth noting that not all green buildings are – and need to be - the same.

Different countries and regions have a variety of characteristics such as distinctive climatic conditions, unique cultures and traditions, diverse building types and ages, or wide-ranging environmental, economic and social priorities – all of which shape their approach to green building.

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